

Prototyping

Low-Fidelity Prototyping

High-Fidelity Prototyping

Prototyping Dimensions

What is a Prototype?

- In other design fields a prototype is a small-scale model:
 - a miniature car
 - a miniature building or town
 - a juicer



What is a Prototype?

- In interaction design it can be (among other things):
 - a series of screen sketches, i.e. not necessarily related
 - a storyboard, i.e. a cartoon-like series of scenes
 - a Powerpoint slide show, e.g., simulation
 - a video simulating the use of a system
 - a cardboard mock-up
 - a piece of software with limited functionality written in a language
- Key point:
 - Goal is to NOT the artifact, but the feedback

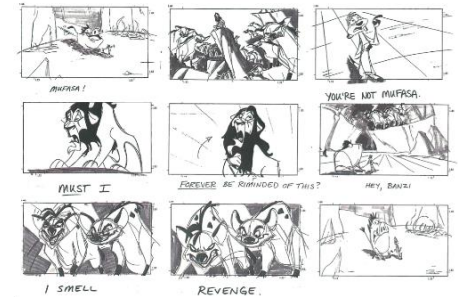
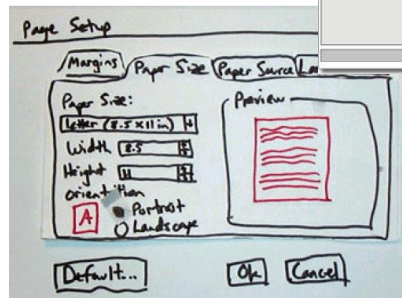


Image source: Nintendo

Prototype Fidelity

Fidelity



Time

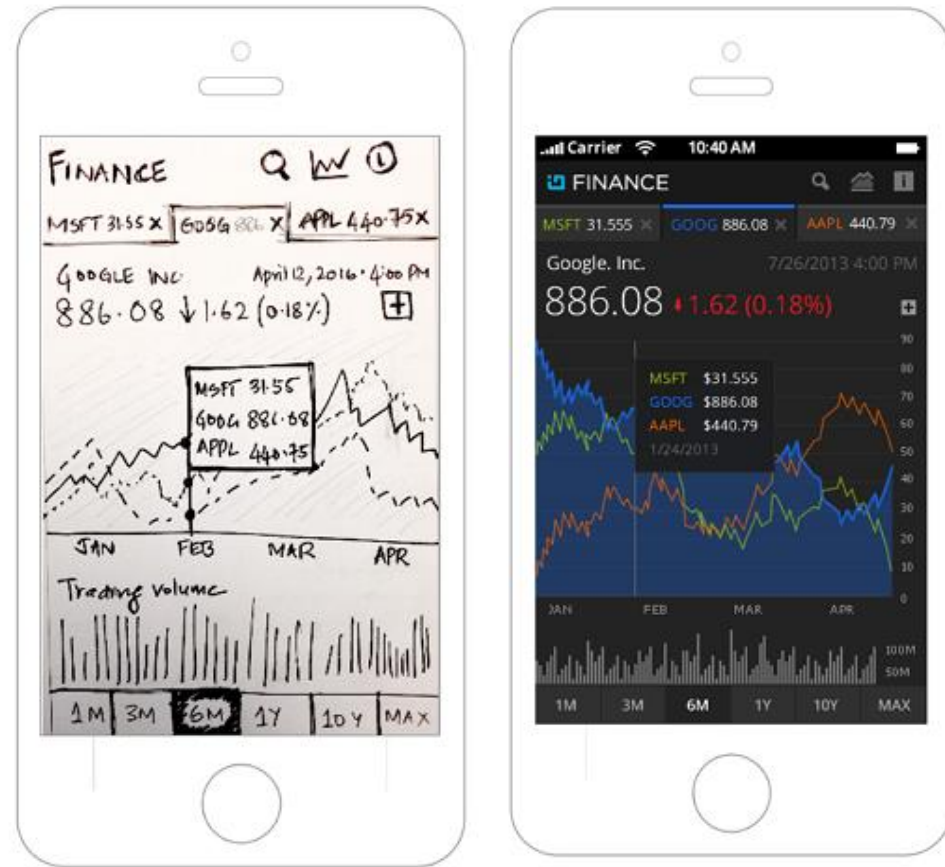
Higher fidelity resembles final product more closely in look and feel, as well as functionality

High-Fidelity Prototypes

- Use hi-fi prototypes, when you have
 - **test the details** of your products in terms of **UI elements**, colour schemes
 - when you want to **test the transitions, animations**, and effects of them on the user and user behaviour
 - want to **know how your target users feel** about your product and you want to get their opinions on your designs.

High-Fidelity Prototypes

- Prototyping with a computer
 - **interactive representation** of the product
 - **closest resemblance to the final design** in terms of details and functionality
 - **simulate some** but not all features of the intended system



What You Can Learn From High-Fidelity Prototypes

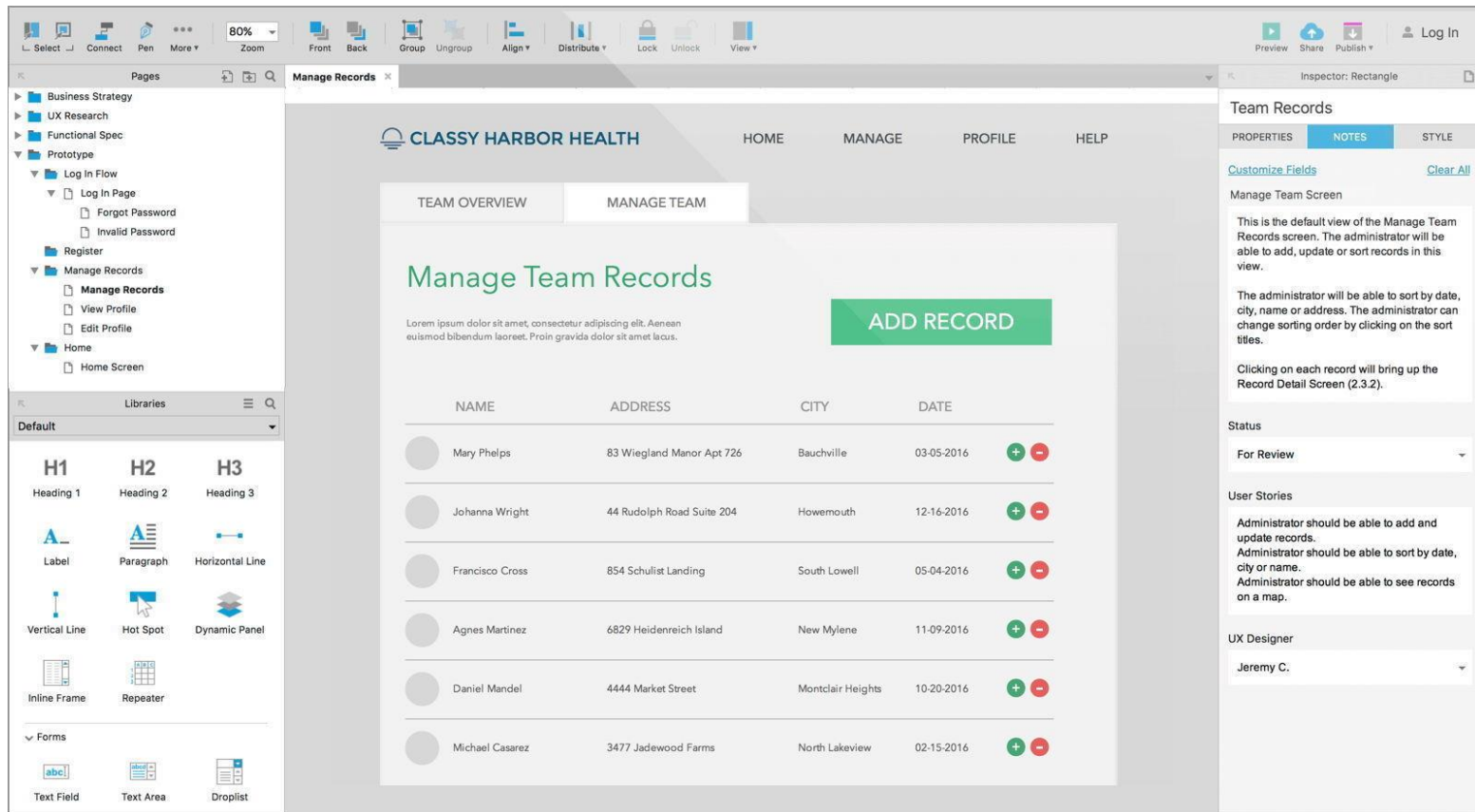
- Everything you learn from a paper prototype, plus:
- Screen layout
 - Is it clear, overwhelming, distracting, complicated?
- Colors, fonts, icons, other elements
 - Well-chosen?
- Interactive feedback
 - Do users notice & respond to status bar messages or cursor changes
- Efficiency issues
 - Scrolling list is too long?

Popular Techniques

- Program it directly in a language
- Use a prototyping tool
 - [Proto.io](#)
 - [InVision](#)
 - [Axure](#)
 - [Principle](#)
 - [Framer](#)
 - [Adobe XD](#)
 - [Sketch](#)

Prototyping Tools

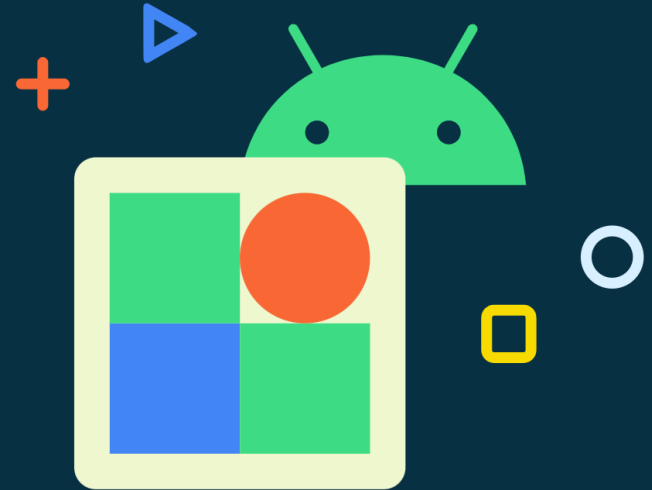
- Axure (<https://www.axure.com/>)
- Free for students (<https://www.axure.com/edu>)



Design Guidelines

Design & Plan

Before you start coding, do your homework to design and plan a great app. Create a great user experience, carefully plan your app architecture, and follow recommended security and privacy practices.



Design guides

GUIDE

Mobile design guides

Get started with Android UI design guides. Follow style guidance, learn about themes, and create a beautiful Android app experience.

GUIDE

Wear design guides

Wear apps have their own unique challenges and design principles. Learn the specifics of designing an Android app for the wrist.

GUIDE

TV design guides

Create a dynamic and elegant Android TV app. Design for TV specs and dimensions and follow our best practices and recommendations.

<https://developer.android.com/design>

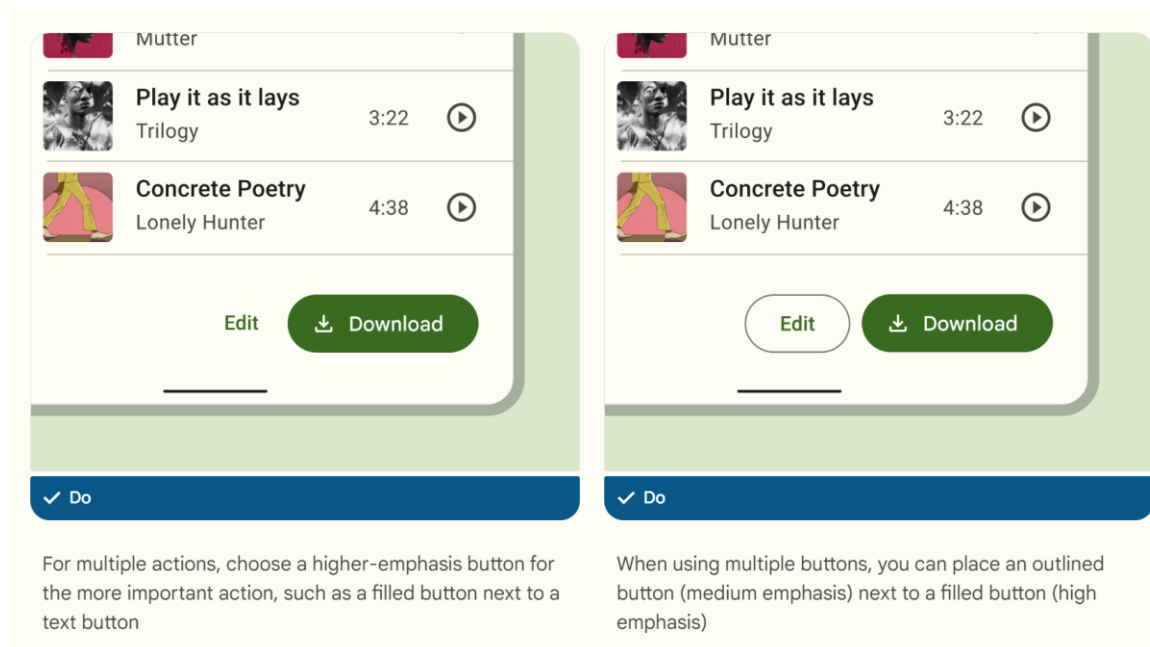
Material Design Guidelines

- **Buttons** (<https://m3.material.io/components/all-buttons>)

- Buttons communicate actions that users can take

Nine types: elevated, filled, filled tonal, outlined, text, icon, segmented, FAB, extended FAB

Each button has either high, medium, or low emphasis



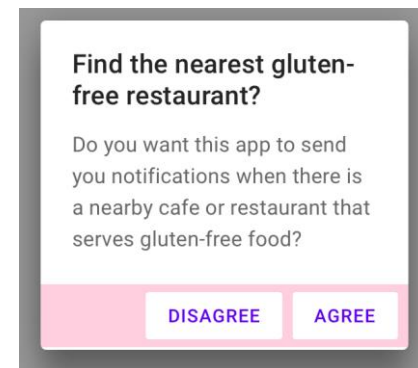
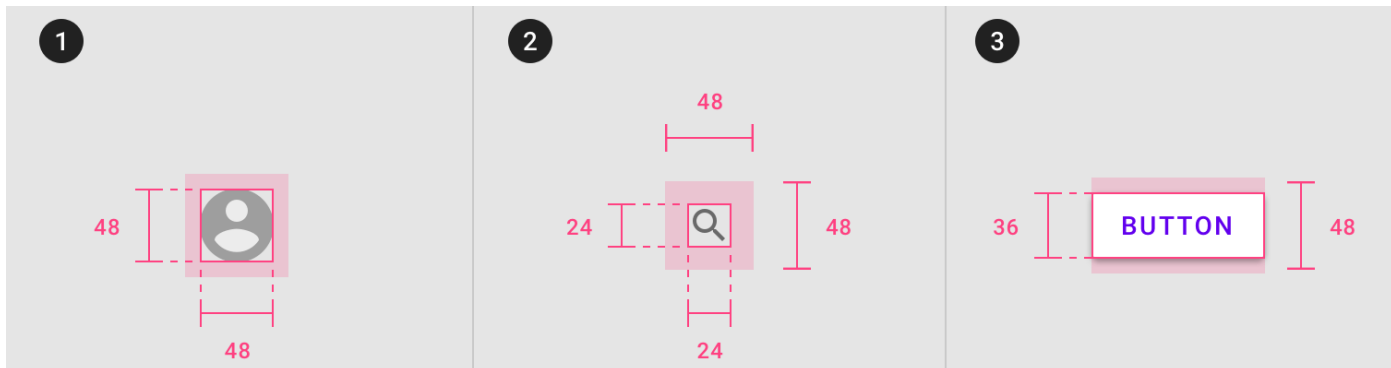
Design Guidelines

▪ Touch targets

- Touch target minimum of 48 x 48 dp
- The recommended target size for touchscreen elements is 7-10mm.
- An icon may appear to be 24 x 24 dp, but the padding surrounding it comprises the full 48 x 48 dp touch target.
- In most cases, touch targets separated by 8dp of space or more promote balanced information density and usability.

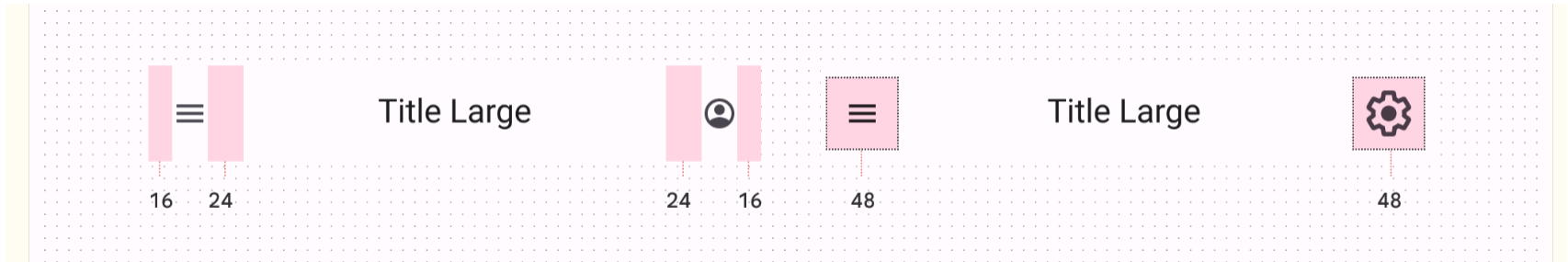
▪ Pointer targets (e.g., mouse pointer)

- Consider making pointer targets at least 44 x 44 dp.

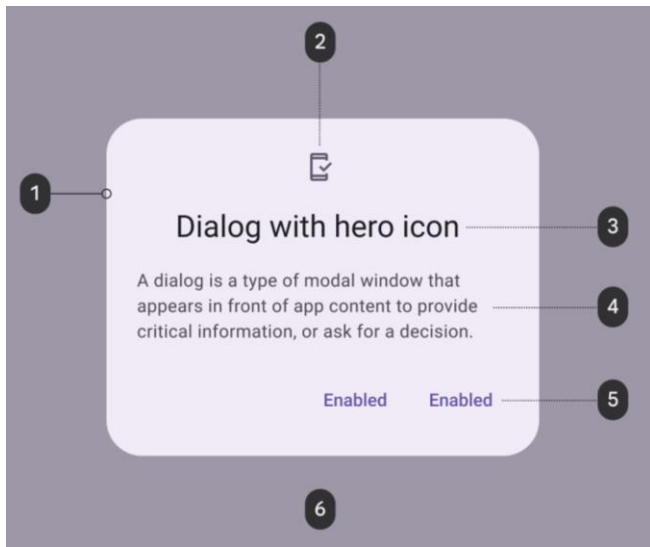


Design Guidelines

▪ Top App bar



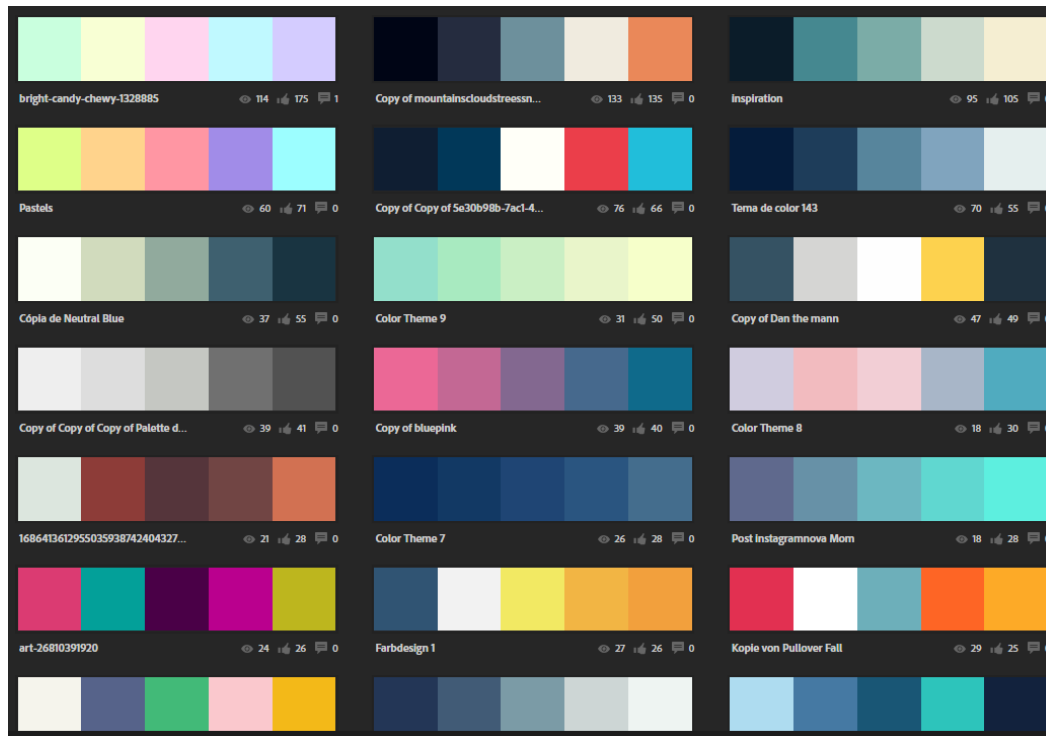
▪ Dialogs



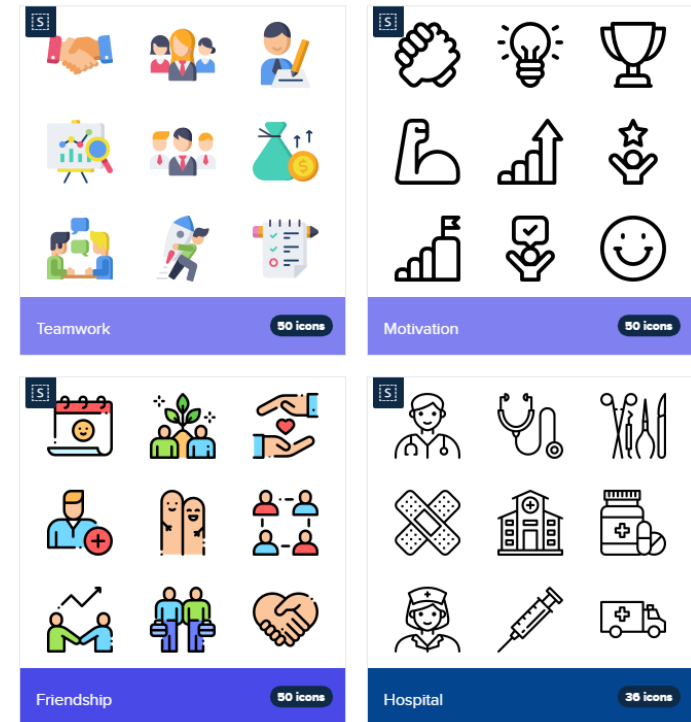
One Line Lists



Colors and Icons



<https://color.adobe.com/create/color-wheel/>

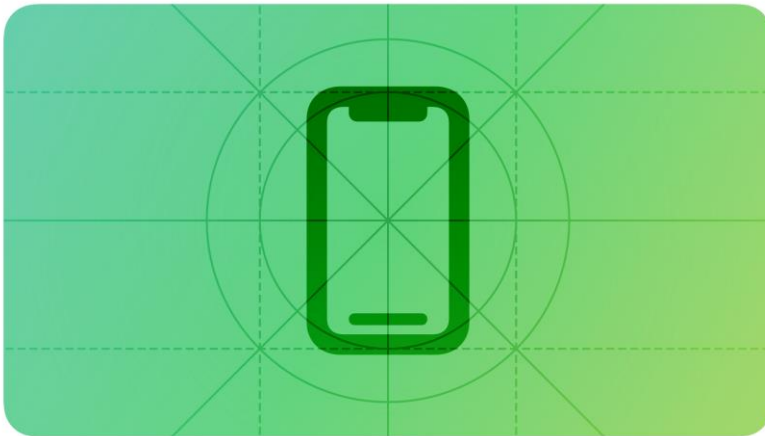


<https://www.flaticon.com/>

Design Guidelines

Designing for iOS

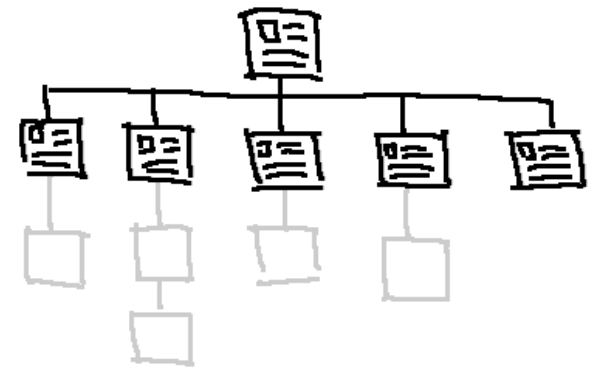
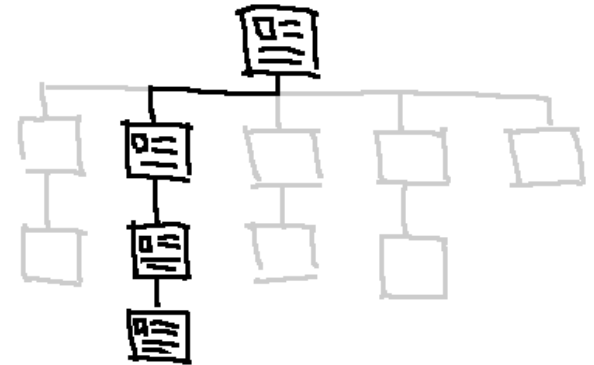
People depend on their iPhone to help them stay connected, play games, view media, accomplish tasks, and track personal data in any location and while on the go.



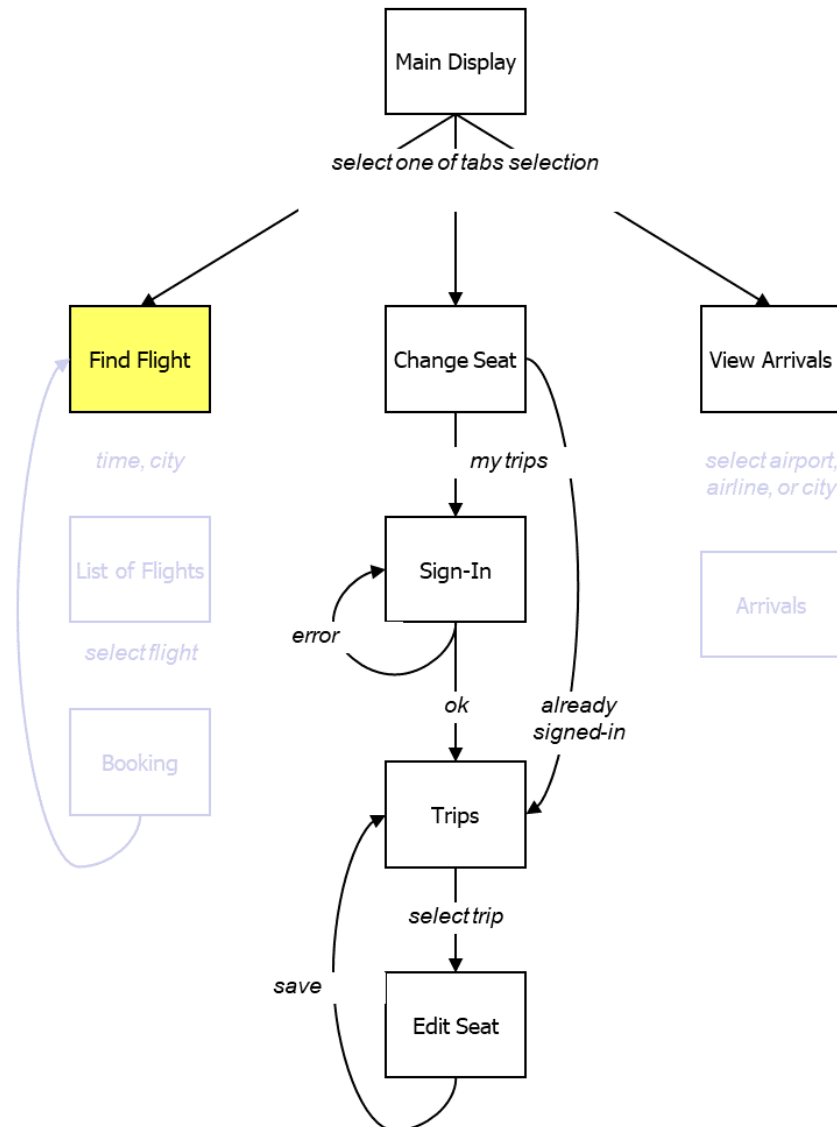
- iOS design Pattern: <https://developer.apple.com/design/human-interface-guidelines/designing-for-ios>
- Web UI Design Patterns
 - <https://www.uxpin.com/studio/ebooks/web-ui-design-patterns-2016-volume-1/>
 - <https://www.uxpin.com/studio/ebooks/web-ui-design-patterns-2016-volume-2/>
- User Interface Design pattern
 - <http://ui-patterns.com/>

Limiting Functionality: How Much to Represent?

- Vertical prototypes
 - includes **in-depth functionality** for only a few selected features
 - common design ideas can be tested in depth
- Horizontal prototypes
 - surface layers includes the **entire user interface** with **limited underlying functionality**



Limiting Functionality: How Much to Represent?



Video Prototypes

- Use video to illustrate how people might **interact with a future technology** in a realistic setting
- Describes relevant **motivations and their activities** over time
- The characters in the video prototype are represented as **personas or characters** drawn from interviews and observations
- Characters placed are in **a particular situation at a particular place** and time

Walkabout Project

- <https://www.youtube.com/watch?v=kWsBvUnvCmg>



Benefits of Video Prototype

- Cheap and fast
- Great communication tools
 - Help achieve common ground
 - Ideally, portable and self-explanatory
- Can server as a 'spec' for developers



How the customer explained it



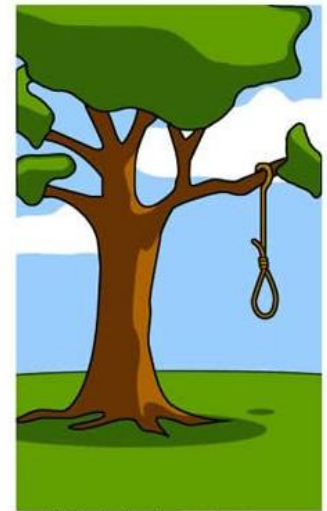
How the project leader understood it



How the analyst designed it



How the programmer wrote it



What the beta testers received

'Wizard-of-Oz' Prototyping

- It's **difficult to prototype** some piece of functionality
- Need: test whether it is actually good before building it!!!
- Solution: **fake it!** Make the interaction as authentic as possible
- Key: user has no idea that the interaction is being faked

Seinfeld - Movie Phone

- <https://www.youtube.com/watch?v=uAb3TcSWu7Q>



Example

- <https://www.youtube.com/watch?v=s9fQxzojS7E>



'Wizard-of-Oz' Prototyping

- The **user thinks they are interacting** with a computer, but a **developer is responding** to output rather than the system.
- “wizard” (sometimes hidden):
 - interprets subjects input according to an algorithm
 - has computer/screen behave in appropriate manner
- Good for:
 - adding **simulated and complex** functionality
 - testing **futuristic ideas**
- Some issues
 - need some means of **controlling system**
 - may need ways to perform **hard computations** quickly
 - must be realistic if simulating **error-prone** technology (e.g., speech recognition)

Low vs. high Fidelity Prototypes

	Advantages	Disadvantages
Low Fidelity	- Fast, low development costs - Evaluate multiple designs - Useful communication device - Useful for identifying requirements - Quick proof-of-concept	- Facilitator-driven - Limited error checking - Poor detailed specifications - Limited use for usability tests
High Fidelity	- Rich functionality - Interactive - User-driven - Clearly defines navigational scheme - Look and feel of product - Marketing and sales tool	- More expensive to develop - More time-consuming

Which Prototyping Method is Best?

- Choose the method that works best for what you are trying to achieve
 - user flow » storyboard
 - evaluate multiple design » paper prototypes
 - overall experience » video prototype
 - functionality » high fed, software/tool
 - ... etc.
- Alternately, maybe you are at different stages in the design. Early on, use techniques that are cheap and quick; later on, use techniques that give a stronger sense of finished idea.

What Did We Cover

- Low-fidelity prototyping
 - Storyboards
 - Paper prototyping
- High-fidelity prototyping
 - Purpose
 - Types
- Vertical (depth) vs. Horizontal (breadth) prototyping
- Low-Fidelity vs. High-Fidelity